

FCT SERVICE NOTE SN06101012A

Class A Service information enclosed.

Date: 10-Aug-06

Subject: LightSpeed 7X “Sherlock” VDAS Detector End-of-Life, Replaced by “Halo” HDAS Detector

Effectivity: LightSpeed 7X VCT 64-Slice Systems

Introduction:

The VCT 64-slice “Sherlock” VDAS (Volara) detector has gone end-of-life and is replaced by the “Halo” HDAS detector. DAS service tools have been updated via software release 06MW29.7 to support the “Halo” HDAS detector, as well as the “Sherlock” VDAS detector. Software release 06MW29.7 also provides the field engineer the support to perform VCT center module replacement.

Purpose:

The purpose of this service note is to introduce the “Halo” HDAS detector to the field service team. The “Halo” HDAS detector is to be cut-into forward production FW39’06, or 25-Sep-06.

Prerequisites:

Software release 06MW29.7, or later, is required for the “Halo” HDAS detector to function. This software release supports both, “Halo” HDAS and “Sherlock” VDAS detectors.

“Sherlock” VDAS detector modules/DAS parts cannot be mixed with “Halo” HDAS detector modules/DAS parts.

“Halo” HDAS detector modules/DAS parts cannot be mixed with “Sherlock” VDAS detector modules.

How to Identify

“Halo” HDAS and “Sherlock” VDAS detectors look physically the same. To determine which hardware to order remotely or on site, look at the Common Service Desktop. The DAStype field indicates which DAS/detector is on the scanner. A VDAS64 or 32 is a “Sherlock” detector and is noted in the parts list as VDAS. The HDAS64 is a HALO detector and is noted in the Parts list as HDAS.

For “Halo,” private DICOM field group 45, tag 1003 may be decoded and displayed to determine the VCT DAS type:

- 17 --> HDAS_32,
- 18 --> HDAS_64.

For “Sherlock,” private DICOM field group 45, tag 1003 may be decoded and displayed to determine the VCT DAS type:

- 14 → VDAS_32,
- 15 → VDAS_64

The "Halo" HDAS detector system is comprised of:

- fifteen (quantity 15) "Halo" DIFB boards and
- fifty-seven (quantity 57) "Halo" detector modules.

The Sherlock "VDAS" detector system is comprised of:

- twenty-nine (quantity 29) "Sherlock" VDAS DIFB boards and
- fifty-seven (quantity 57) "Sherlock" VDAS detector modules.

For "Halo," half (50%) of the DIFB boards have been designed-out by the "Halo" DAS design because one "Halo" DIFB board drives four (4) detector modules (A and B sides) from power supply and data collection standpoints.

The two DAS/detector subsystems look the same from an outside perspective.

Training / Product Features:

General Background

The "Halo" HDAS 64-slice VCT detector improves quality and reduces cost by requiring fewer parts, fewer interconnects, and less power than its predecessor, compared to the "Sherlock" VDAS detector.

Service DAS tools supporting the HDAS detector have been updated to support both VDAS and HDAS.

Thermal Error After Hardware Reset

A detector or plenum thermal error may be seen right after a hardware reset. This is due to a timing issue in firmware. The condition clears quickly. The issue may be seen due to filtered average temperature readings that have not had a time to collect enough data for the average to be within specifications before firmware tries to report the average value. Don't troubleshoot this unless the condition continues for more than a few minutes.

Scan Analysis

Reminder for all software releases. Must pay attention to the number of channels displayed by scan analysis when analyzing scan and calibration data. If the displayed data is showing 912 channels, then the mapping of display to actual DAS channels is one to one. This is currently true for MSD and VVC scan data displays. If the number of channels is less than 912, then the reference channels are not being displayed so you need to add 8 to the channel displayed to match the physical DAS channels. This offset can be seen in the calibration data plots for a scan. The key is to look at the total channels displayed to know if there is an offset.

Error Message -- Misleading Detector Channel Label

The following error 2300200003 is not showing the correct channel, it is not including the first 8 reference channels and is starting at zero. So instead of channel 424, the correct channel is actually $424 + 8(\text{ref channels}) + 1(\text{zero offset}) = \text{channel } 433$. For these calibration error messages, this is the same issue in 16 slice. All other error messages that indicate module number not just channel are correct as displayed.

Mon Jul 31 22:23:16 2006

Error: 230020003

Host: ct64_309

Process: accs

File: cal_proc_pb.c

Line: 437

Bad channel found at row: 14, channel: 424.

DAS/Detector Power Supply and Thermal Data

The DASTools Aux channels functionality and Scan Data Path Power Supplies test have been improved to label the power supplies by DIFB and Digital Module (FRDM) they apply to. DASTools Aux channel results log will only show failures just as all the other DASTools tests. The full list of results will still be found in the DASTools Aux channel history log, viewable from the Software History Logs viewer on the CSD just like all the other test data.

Service Toolkit:

For Class C/M customers / customer sites, software release 06MW29.7, p/n 5182512, requires the following Class C and M software:

- 06MW29.7 Class C p/n 5192235, distributed per TAB #182 distribution list
- 06MW29.7 Class M p/n 5181129, distributed per TAB #182 distribution list

Service Docs:

Service documentation has been updated to support the “Halo” HDAS detector, and can be found on the VCT Service Information CDROM, p/n 5116423-200, Rev 11 (or later). This VCT service information CDROM is found inside technical publication collector, p/n B7864HJ.

FRUs:

The following field replaceable units (FRUs) have been created to support the “Halo” HDAS detector:

P/N “Halo” HDAS FRU	Part Description	Qty / System	Replaces P/N
5143018	DIFB PWA, Digital Interface Board	15	N / A
5196401	Center Detector Module Kit, HDAS (slots 27-31)	5	N / A
5196402	Outer Detector Module Kit, HDAS (slots 1-26,32-57)	52	N / A
5130202-2	Low Channel Backplane, “Halo” HDAS	1	N / A
5130204-2	High Channel Backplane, “Halo” HDAS	1	N / A
5166833	DIFB Cover Assembly	2	N / A
5170168	Sub-D Connector Cover - VCT Plenum	2	N / A

“Halo” HDAS and “Sherlock” VDAS FRUs cannot be mixed.

IMPORTANT: Reminder to reference Service Note SN05091210c, or later, defining any special service actions for center module replacements.

Planned Maintenance:

No changes to planned maintenance schedules; no additions.

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